

BGFNET: SUPPLEMENTARY MATERIALS

1. EVALUATION METRICS

We adopt six mainstream evaluation metrics to comprehensively assess the segmentation performance of the proposed model, including mean Dice coefficient (mDic), mean Intersection over Union (mIoU), mean F-measure (F_β^m), Structure-measure (S_α), maximum Enhanced-alignment measure ($\max E_m$), and Mean Absolute Error (MAE).

Among them, mDic and mIoU are region-level similarity metrics, which focus on measuring the overall overlap consistency between the predicted segmentation mask and the ground truth, reflecting the model’s ability to capture the complete region of the target object. MAE is a pixel-level error metric, which calculates the average absolute difference between the predicted probability map and the binary ground truth, and is sensitive to subtle pixel-level prediction deviations. F_β^m comprehensively balances precision and recall, and is obtained by averaging the F-measure values across all threshold settings, providing a robust evaluation of the model’s comprehensive classification performance. S_α focuses on the structural similarity between the prediction and the ground truth, evaluating the segmentation quality from both region-level and object-level structural perspectives. $\max E_m$ evaluates the segmentation results from both pixel-level local alignment and image-level global consistency, and its maximum value reflects the optimal matching degree between the predicted result and the ground truth. All evaluation results are generated using the standard medical image segmentation evaluation toolbox available from PraNet.

2. DATASETS AND DATA SPLITS

We use Kvasir-SEG and CVC-ClinicDB as the two source domains. The source training set contains 900 Kvasir-SEG images and 550 CVC-ClinicDB images. We independently reserve 10% of the samples from each source domain for validation and use the remaining 90% for training. Accordingly, the training set contains 810 Kvasir-SEG and 495 CVC-ClinicDB images, while the validation set contains 90 and 55 images, respectively. The checkpoint with the highest domain-balanced validation Dice score is selected for evaluation.

The selected model is evaluated on five public benchmarks: Kvasir-SEG, CVC-ClinicDB, CVC-ColonDB, CVC-300, and ETIS-LaribPolypDB. CVC-ColonDB, CVC-300,

and ETIS-LaribPolypDB are not used during training or validation and are treated as unseen domains for evaluating cross-domain generalization.

3. LOSS FUNCTION

BGF-Net is trained with a multi-output segmentation loss and three background-aware auxiliary objectives. The segmentation loss supervises the final prediction P_1 and the auxiliary predictions P_2 , P_3 , and P_4 . The auxiliary objectives supervise foreground-background allocation, background semantic prediction, and cross-domain background feature alignment, respectively.

3.1. Multi-Output Segmentation Loss

Let Y denote the ground-truth foreground mask. Following the weighted structure loss, a pixel-wise weight map is first computed as

$$W = 1 + 5 |\text{AvgPool}_{31}(Y) - Y|, \quad (1)$$

where $\text{AvgPool}_{31}(\cdot)$ denotes average pooling with a 31×31 kernel. The weight map assigns larger weights to pixels around lesion boundaries.

For a prediction logit P , its probability map is denoted by $\hat{Y} = \sigma(P)$. The weighted binary cross-entropy loss and weighted IoU loss are defined as

$$\mathcal{L}_{\text{wbce}} = \frac{\sum_p W^{(p)} \text{BCE}(P^{(p)}, Y^{(p)})}{\sum_p W^{(p)}}, \quad (2)$$

$$\mathcal{L}_{\text{wiou}} = 1 - \frac{\sum_p W^{(p)} \hat{Y}^{(p)} Y^{(p)} + 1}{\sum_p W^{(p)} (\hat{Y}^{(p)} + Y^{(p)} - \hat{Y}^{(p)} Y^{(p)}) + 1}. \quad (3)$$

The structure loss is given by

$$\mathcal{L}_{\text{str}}(P, Y) = \mathcal{L}_{\text{wbce}} + \mathcal{L}_{\text{wiou}}. \quad (4)$$

The multi-output segmentation loss is formulated as

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{str}}(P_1, Y) + \lambda_{\text{aux}} \sum_{i=2}^4 \mathcal{L}_{\text{str}}(P_i, Y), \quad (5)$$

where λ_{aux} controls the contribution of the auxiliary decoder predictions and is set to 0.2 by default. All predictions are resized to the input-image resolution before computing $\mathcal{L}_{\text{seg}}$.

3.2. Reliable Foreground–Background Supervision

Direct supervision near polyp boundaries can be unreliable because of annotation ambiguity and severe foreground–background similarity. Therefore, we construct reliable foreground and background regions by eroding and dilating the ground-truth mask:

$$\begin{aligned} R^{\text{fg}} &= \text{Erode}(Y, r), \\ R^{\text{bg}} &= 1 - \text{Dilate}(Y, r), \end{aligned} \quad (6)$$

where $\text{Erode}(\cdot, r)$ and $\text{Dilate}(\cdot, r)$ denote morphological erosion and dilation with radius $r = 5$, respectively. The valid supervision region is defined as

$$R^{\text{valid}} = \text{clip}(R^{\text{fg}} + R^{\text{bg}}, 0, 1), \quad (7)$$

where $\text{clip}(\cdot, 0, 1)$ truncates its input to the interval $[0, 1]$. Consequently, pixels within the uncertain boundary band are excluded from the background-aware supervision.

At each BGA stage, the allocation masks M_i^{fg} and M_i^{bg} are supervised within the reliable region. All masks are resized to the corresponding feature resolution by nearest-neighbor interpolation. The allocation loss is defined as

$$\mathcal{L}_{\text{mask}} = -\frac{1}{3} \sum_{i=2}^4 \frac{\sum_p R_i^{\text{valid},(p)} \left[Y_i^{(p)} \log M_i^{\text{fg},(p)} + (1 - Y_i^{(p)}) \log M_i^{\text{bg},(p)} \right]}{\sum_p R_i^{\text{valid},(p)} + \varepsilon}. \quad (8)$$

where $i = 2, 3, 4$ indexes the BGA stages, p indexes spatial locations, $\varepsilon = 10^{-6}$ is a numerical constant, and Y_i and R_i^{valid} denote the ground-truth mask and valid-region mask resized to the resolution of stage i , respectively.

3.3. Background Semantic Supervision

At each BGA stage, the background feature B_i is mapped to a background logit G_i^{bg} , and the corresponding background probability map is obtained by

$$Q_i^{\text{bg}} = \sigma(G_i^{\text{bg}}). \quad (9)$$

The background branch is supervised using the background target $1 - Y_i$ within the reliable region:

$$\begin{aligned} \mathcal{L}_{\text{bg-sem}} &= \frac{1}{3} \sum_{i=2}^4 \left[\mathcal{L}_{\text{bce}}(G_i^{\text{bg}}, 1 - Y_i, R_i^{\text{valid}}) \right. \\ &\quad \left. + \lambda_{\text{dice}} \mathcal{L}_{\text{dice}}(Q_i^{\text{bg}}, 1 - Y_i, R_i^{\text{valid}}) \right]. \end{aligned} \quad (10)$$

where $i = 2, 3, 4$ indexes the BGA stages, G_i^{bg} denotes the background logit, and Q_i^{bg} denotes the corresponding background probability map. The functions $\mathcal{L}_{\text{bce}}(\cdot)$ and $\mathcal{L}_{\text{dice}}(\cdot)$ denote binary cross-entropy with logits and Dice loss computed only over valid pixels, respectively. We set $\lambda_{\text{dice}} = 1.0$.

This objective encourages the background branch to capture stable non-polyp structures while avoiding ambiguous boundary regions.

3.4. Background Invariance Loss

To improve robustness under domain shifts, BGF-Net aligns reliable background representations from the third and fourth BGA stages. For each sample from source domain d , the background feature is pooled over the reliable background region and then ℓ_2 normalized:

$$\mathbf{z}_{i,n}^d = \text{Norm}\left(\text{Pool}\left(B_{i,n}^d, R_{i,n}^{\text{bg},d}\right)\right), \quad (11)$$

where i indexes the BGA stage, n indexes a sample from source domain d , $R_{i,n}^{\text{bg},d}$ denotes the reliable background mask resized to the resolution of $B_{i,n}^d$, $\text{Pool}(\cdot)$ denotes masked global average pooling, and $\text{Norm}(\cdot)$ denotes ℓ_2 normalization. Let $\mathcal{Z}_i^d$ denote the set of normalized background vectors from source domain d at stage i .

For the Kvasir-SEG and CVC-ClinicDB source domains, the background invariance loss is calculated using multi-kernel MMD:

$$\mathcal{L}_{\text{inv}} = \frac{1}{|\mathcal{S}_{\text{valid}}|} \sum_{i \in \mathcal{S}_{\text{valid}}} \text{MMD}_{\text{MK}}^2(\mathcal{Z}_i^a, \mathcal{Z}_i^b), \quad (12)$$

where a and b denote the two source domains, and $\mathcal{S}_{\text{valid}}$ contains valid stages selected from the third and fourth BGA stages. A stage is included only when both domains contain at least two samples. The multi-kernel MMD is computed by averaging Gaussian kernels with bandwidths $\Sigma = \{0.5, 1.0, 2.0, 4.0\}$. This loss encourages background representations from different domains to share similar feature distributions while preserving foreground discrimination through the segmentation objective.

3.5. Overall Objective

The main training objective is

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\text{seg}} + \lambda_{\text{bg}} (\mathcal{L}_{\text{mask}} + \mathcal{L}_{\text{bg-sem}}) \\ &\quad + \lambda_{\text{inv}} \mathcal{L}_{\text{inv}}, \end{aligned} \quad (13)$$

where λ_{bg} jointly controls the contributions of the foreground–background allocation loss and background semantic loss, while λ_{inv} controls the contribution of the background invariance loss. Unless otherwise stated, we set $\lambda_{\text{bg}} = 0.60$ and $\lambda_{\text{inv}} = 0.20$.

Table 1. Detailed comparison with SOTA methods on the Kvasir-SEG and CVC-ClinicDB datasets. The best and second-best available results are highlighted in **bold** and underlined, respectively. EFA-Net did not report max E_m or MAE .

Method	Kvasir-SEG						CVC-ClinicDB					
	$mDic$	$mIoU$	F_β^w	S_α	max E_m	MAE	$mDic$	$mIoU$	F_β^w	S_α	max E_m	MAE
UNet (2015)	0.818	0.746	0.794	0.858	0.893	0.055	0.823	0.755	0.811	0.889	0.954	0.019
PraNet (2020)	0.896	0.838	0.880	0.913	0.944	0.030	0.900	0.849	0.896	0.937	0.979	0.009
Polyp-PVT (2021)	<u>0.918</u>	<u>0.868</u>	0.915	0.924	<u>0.970</u>	0.021	<u>0.935</u>	<u>0.885</u>	<u>0.935</u>	0.942	0.989	<u>0.007</u>
EFA-Net (2025)	0.914	0.861	0.906	<u>0.929</u>	–	–	0.919	0.871	0.916	0.943	–	–
FMCA-Net (2025)	0.906	0.851	0.902	0.917	0.962	0.026	0.930	0.881	0.928	<u>0.946</u>	<u>0.992</u>	<u>0.007</u>
PraNet-V2 (2026)	0.915	0.861	0.904	0.925	0.960	0.023	0.931	0.881	0.928	0.944	0.986	0.008
BGF-Net (Ours)	0.921	0.870	<u>0.914</u>	0.936	0.982	<u>0.022</u>	0.938	0.891	0.937	0.957	0.995	0.006

Table 2. Detailed comparison with SOTA methods on the CVC-ColonDB, CVC-300, and ETIS datasets. The best and second-best available results are highlighted in **bold** and underlined, respectively. EFA-Net did not report max E_m or MAE .

Method	CVC-ColonDB						CVC-300						ETIS					
	$mDic$	$mIoU$	F_β^w	S_α	max E_m	MAE	$mDic$	$mIoU$	F_β^w	S_α	max E_m	MAE	$mDic$	$mIoU$	F_β^w	S_α	max E_m	MAE
UNet (2015)	0.512	0.444	0.498	0.712	0.776	0.061	0.710	0.627	0.684	0.843	0.875	0.022	0.398	0.335	0.366	0.684	0.740	0.036
PraNet (2020)	0.713	0.642	0.699	0.820	0.872	0.043	0.874	0.799	0.843	0.925	0.972	0.010	0.630	0.568	0.601	0.793	0.841	0.031
Polyp-PVT (2021)	0.793	0.711	0.780	0.858	0.908	0.032	0.893	0.825	<u>0.878</u>	0.930	0.976	0.009	0.786	0.703	<u>0.753</u>	0.871	0.915	0.018
EFA-Net (2025)	0.774	0.696	0.753	0.855	–	–	0.894	<u>0.830</u>	<u>0.878</u>	0.941	–	–	0.749	0.670	0.698	0.858	–	–
FMCA-Net (2025)	<u>0.795</u>	<u>0.714</u>	<u>0.782</u>	<u>0.861</u>	<u>0.921</u>	0.032	0.892	0.827	0.872	0.933	0.978	<u>0.008</u>	<u>0.791</u>	<u>0.712</u>	<u>0.753</u>	<u>0.877</u>	<u>0.936</u>	0.017
PraNet-V2 (2026)	0.767	0.687	0.752	0.845	0.892	0.036	0.899	0.831	0.885	<u>0.940</u>	0.976	0.007	0.764	0.687	0.730	0.865	0.888	<u>0.015</u>
BGF-Net (Ours)	0.802	0.726	0.787	0.869	0.951	0.032	<u>0.896</u>	0.827	0.874	0.937	0.988	0.007	0.834	0.761	0.806	0.908	0.968	0.012

Table 3. Sensitivity to the reliable-region radius r under $\lambda_{bg} = 0.60$ and $\lambda_{inv} = 0.20$. Avg. denotes the average over five datasets.

r	Kvasir-SEG		CVC-ClinicDB		CVC-ColonDB		CVC-300		ETIS		Avg.	
	$mDic$	$mIoU$	$mDic$	$mIoU$	$mDic$	$mIoU$	$mDic$	$mIoU$	$mDic$	$mIoU$	$mDic$	$mIoU$
0	0.918	0.871	0.927	0.881	0.810	0.730	0.885	0.817	0.823	0.753	0.873	0.810
3	0.921	0.870	0.938	0.891	0.802	0.726	0.896	0.827	0.834	0.761	0.878	0.815
5	0.922	0.873	0.922	0.875	0.802	0.726	0.874	0.806	0.821	0.751	0.868	0.806
7	0.913	0.862	0.938	0.893	0.786	0.713	0.897	0.831	0.775	0.708	0.862	0.801
9	0.918	0.866	0.942	0.898	0.799	0.720	0.870	0.802	0.815	0.734	0.869	0.804

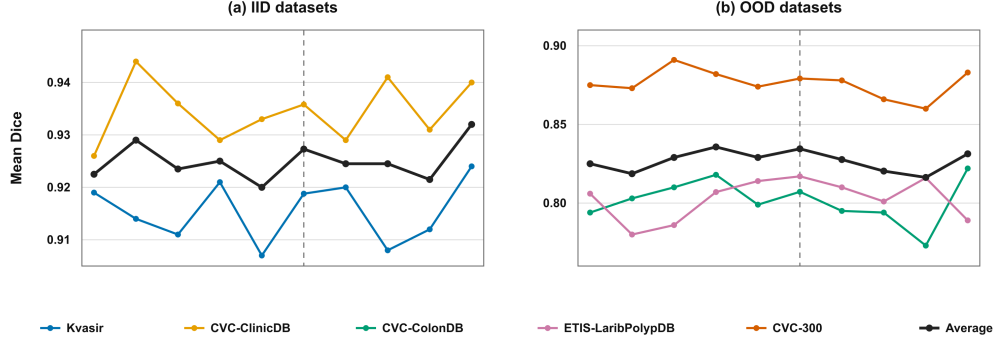


Fig. 1. Sensitivity analysis of the background supervision weight λ_{bg} , where $\lambda_{mask} = \lambda_{sem} = \lambda_{bg}$. From left to right, the horizontal-axis points correspond to $\lambda_{bg} \in \{0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, 1.00\}$. The dashed line indicates the selected setting $\lambda_{bg} = 0.60$.

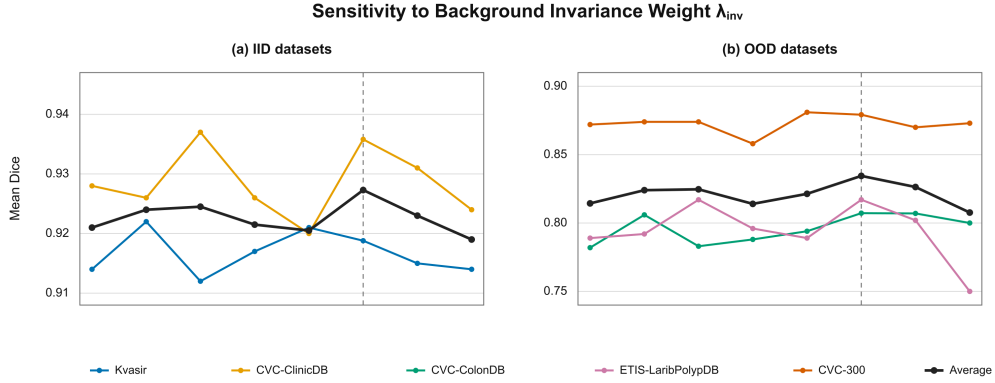


Fig. 2. Sensitivity analysis of the background-invariance weight λ_{inv} for MMD regularization. From left to right, the horizontal-axis points correspond to $\lambda_{inv} \in \{0, 0.03, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$. The dashed line indicates the selected setting $\lambda_{inv} = 0.20$.

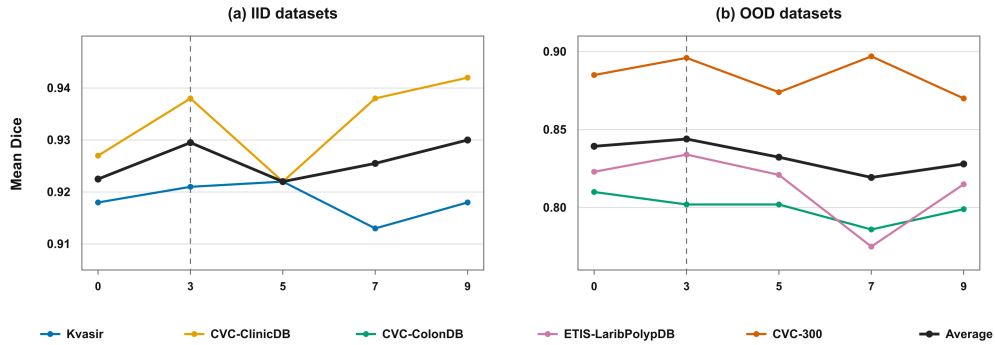


Fig. 3. Sensitivity analysis of the reliable-region radius r under $\lambda_{bg} = 0.60$ and $\lambda_{inv} = 0.20$. From left to right, the horizontal-axis points correspond to $r \in \{0, 3, 5, 7, 9\}$. The dashed line indicates the selected setting $r = 3$.